ARUN ROSHAN A S

Automotive Systems & R&D Engineer | ECU & Automotive Electronics | Powertrain | Vehicle Architecture | Alternative Fuels

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PROFESSIONAL SUMMARY

Automotive Systems & R&D Engineer with 2 years of founder-led engineering experience spanning ECU and automotive electronics, powertrain, internal combustion engines, vehicle architecture, and alternative-fuel technologies. Cross-disciplinary strength across electrical, electronic, mechanical, and control-systems domains, with hands-on delivery in engine-fuel integration, sensing, and closed-loop controls. Proven track record engaging national regulatory and testing bodies (ARAI, PESO) and international safety certifiers (TÜV SÜD, Bureau Veritas) on hydrogen retrofit technology. Seeking to bring this multidisciplinary, results-driven profile to international automotive R&D, powertrain, and alternative-fuel programs at leading global OEMs.

EDUCATION

B.E. Electrical & Electronics Engineering (Minor: Mechanical Engineering) *2023 – 2027*

KIT – Kalaignarkarunanidhi Institute of Technology, Coimbatore, India | CGPA: 7.70 / 10.0

Relevant Coursework: Automotive Powertrain Engineering, Automotive Electronics & ECU Systems, Control Systems & Instrumentation, Thermodynamics, Fluid Mechanics, Power Electronics, Automotive Sensors & Actuators.

PROFESSIONAL EXPERIENCE

R&D Head & Co-Founder | Herrscher Mobility *2024 – Present*

- Co-founded and led R&D for a deep-tech automotive venture retrofitting existing ICE platforms with hydrogen and alternative-fuel systems, spanning ICE retrofit architecture and regulatory review.
- Developed a hydrogen-fuelled six-stroke ICE architecture targeting **>22% thermal efficiency** through optimized combustion strategy, fuel integration, engine design stability, and operating parameters.
- Optimized a hydrogen-blend ICE retrofit architecture for existing heavy commercial vehicle platforms, integrating fuel injection and ECU control to deliver **18% higher power and torque output** (*ARAI-reviewed*).
- Architected ECU and automotive electronics systems for hydrogen powertrains, integrating sensor feedback, fuel injection, and actuator control to establish a unified control layer across Herrscher's engine platforms.

Project Intern – Powertrain & Electrical Systems | Sona BLW Precision Forgings Ltd *2025 – march*

- Joined to develop drivetrain architecture for a hydrogen-hybrid powertrain, evaluating drive-motor and hub-motor systems for integration into the hybrid drive layout.
- Supported motor production, PLC-based automation, and robotics processes across electrical-mechanical manufacturing operations relevant to hydrogen-hybrid drivetrain development.

INDEPENDENT RESEARCH

- Designed a novel compression spark-assisted ignition (CSI) combustion strategy combining compression- and spark-ignition principles, alongside a self-designed gaseous direct-injection (GDI) system for alternative fuels. (2025)
- Applied the CSI/GDI architecture to four-stroke ammonia combustion, designing and building a custom ECU control strategy with EGR and dedicated safety protocols; reviewed by both **ARAI and PESO**. (2026)
- Simulated ammonia GDI combustion with hydrogen pilot ignition, achieving **12% higher torque than diesel**, while evaluating injection-ignition efficiency trade-offs. (2026)
- Built a deep-learning-based adaptive ECU control system for existing diesel HCV platforms using a Google Coral edge-AI model to process GPS and load-condition data and optimize injector timing in real time, achieving a simulated **12% fuel-efficiency improvement**. (2024)

TECHNICAL SKILLS

- **Mechanical & Powertrain:** IC Engines, Combustion Systems, Engine Retrofit Architecture, Powertrain Integration, CFD Analysis
- **Electronics & Controls:** ECU Development, Sensors & Actuators, Fuel Injection & Ignition Systems, Control Systems, Electric Drives, AI-Assisted ECU Optimization
- **Embedded & Communication Protocols:** SPI, I2C, CAN, AUTOSAR (Layered Architecture)
- **Programming:** C, Python, Java
- **Design & Simulation Tools:** CAD, MATLAB/Simulink, ANSYS
- **AI & Machine Learning:** Applied AI, Deep Learning, Edge AI (Google Coral)
- **Fluid Power & Manufacturing:** Hydraulics, Pneumatics, Industrial Manufacturing Processes

ACHIEVEMENTS

- Secured multi-body regulatory engagement for hydrogen retrofit technology, including concept review and technical mentorship from **ARAI**, alongside active engagement with **TÜV SÜD and Bureau Veritas**.
- Engaged directly with international hydrogen industry leaders, including H2 Innovations and Fluitron USA, on metal hydride storage, safety certification, and commercialization pathways.
- Ranked among the **Top 15 innovators nationally** in the IIT Delhi Ideathon. (2025)
- Collaborated with the **CSIR-NIIST** scientific team to develop a hydrogen fuel blend for H_2 retrofit and support PESO review.